

Introduction to Data Analytics

Your First Step Into the World of Data

■ FOUNDATIONS

■ LEARNING OUTCOME

By the end of this PDF, you will understand what data analytics is, why it matters in business and careers, and the core workflow every analyst follows.

■ Skills You Will Gain

- Define data analytics and its types
- Understand the analyst workflow (OSEMN)
- Know real-world use cases
- Identify career paths in data
- Understand why data analytics is high-paying

■ Data Analytics professionals earn ■4–30 LPA in India and \$60K–\$120K globally. This skill is #1 in demand for 2024–2025.

SECTION 1

What is Data Analytics?

Data Analytics is the science of examining raw data to draw meaningful conclusions, discover patterns, and support decision-making. It transforms **raw numbers into actionable insights**.

Type	Description & Example
Descriptive	What happened? → Monthly sales report showing ■5L revenue
Diagnostic	Why did it happen? → Sales dropped because of a competitor's discount
Predictive	What will happen? → ML model predicts next month's demand
Prescriptive	What should we do? → Recommend restocking 200 units of Product X

SECTION 2

The Analytics Workflow (OSEMN)

1	Obtain Collect data from databases, APIs, surveys, web scraping, or CSV files.
2	Scrub Clean data: remove duplicates, handle missing values, fix formats.
3	Explore Run EDA: descriptive stats, distributions, correlations.
4	Model Apply statistical or ML models to find patterns and predictions.
5	iNterpret Translate results into business language and actionable insights.

SECTION 3

Real-World Use Cases

Industry	Analytics Use Case
E-Commerce	Amazon uses analytics to recommend products (35% of revenue)
Healthcare	Predict patient readmission, optimize drug dosage
Finance	Fraud detection — Visa blocks 150M+ fraudulent txns/day

Sports	IPL teams use data to pick players and strategy
Education	Track student performance and predict dropout risk

SECTION 4

Tools Every Analyst Uses

Tool	Purpose
Excel / Google Sheets	Quick analysis, pivot tables, basic charts
SQL	Query databases, join tables, filter large datasets
Python (Pandas, NumPy)	Advanced analysis, automation, ML
Power BI / Tableau	Interactive dashboards and visual reports
Statistics	Hypothesis testing, probability, regression

SECTION 5

Career Roadmap

1	Level 1: Analyst Basics Excel, SQL, basic stats → Junior Data Analyst
2	Level 2: Visualization Power BI / Tableau → Business Intelligence Analyst
3	Level 3: Python + Stats Pandas, NumPy, regression → Data Analyst
4	Level 4: ML Basics Scikit-learn, prediction models → Data Scientist
5	Level 5: Specialization NLP / Deep Learning / MLOps → Senior Data Scientist

PRACTICE QUESTIONS

- Q1. What is the difference between Descriptive and Predictive Analytics?

Answer: Descriptive analyzes past data (what happened); Predictive uses models to forecast future events.
- Q2. Name 3 tools a Data Analyst uses daily.

Answer: Excel, SQL, Power BI (or Python, Tableau).
- Q3. What does OSEMN stand for?

Answer: Obtain, Scrub, Explore, Model, iNterpret.
- Q4. Give one real-world example of analytics in business.

Answer: Netflix uses viewing history data to recommend shows, reducing churn by 30%.

Q5. Why is data analytics considered a high-paying skill?

Answer: Every company generates data and needs people who can turn it into decisions. Demand far exceeds supply.

■ MINI PROJECT

Apply what you've learned with a real dataset!

Dataset: Monthly Sales CSV (12 months, 5 products)

Steps:

1	Load Open CSV in Excel or Python (<code>pd.read_csv</code>)
2	Explore How many rows? Any missing values?
3	Summarize Find total, average, min, max sales
4	Visualize Bar chart: Sales by Month
5	Conclude Which month had highest sales? Why?

Expected Output: A bar chart + summary table showing monthly trends

Insights to Find: Peak sales month, lowest sales month, average monthly revenue

■ BONUS RESOURCES

Free Dataset	kaggle.com/datasets — search 'sales data'
Free Tool	Google Colab — Python in browser, free GPU
Learning	Google Data Analytics Certificate (Coursera, free audit)
Community	r/datascience, Analytics Vidhya, Towards Data Science
Book	'Storytelling with Data' by Cole Nussbaumer Knaflic

■ Key Concepts Cheat Sheet

Data Analytics	Turning raw data into insights
Descriptive	Summarize past — avg, sum, count
Diagnostic	Find root cause — drill down
Predictive	Forecast future — ML models
Prescriptive	Recommend action — optimization
OSEMN	Obtain → Scrub → Explore → Model → iNterpret
KPI	Key Performance Indicator — measurable goal

